

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2026

SP4 - Pensions and Other Benefits

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) ABC Manufacturing Ltd., a private sector company with 1,200 employees, currently operates a Defined Benefit (DB) final salary pension scheme in addition to statutory benefits under the Employees' Provident Fund.

Due to increasing funding volatility and rising longevity, the company decides to close the DB scheme to future accrual and introduce a Defined Contribution (DC) scheme regulated by the Pension Fund Regulator. To evaluate the scheme design, two employees are considered:

- **Employee A:** Age 58, retiring in 2 years, 30 years of service (majority benefits under DB).
- **Employee B:** Age 28, recently joined, entire future benefits under DC.

i) List four key stakeholders in ABC Manufacturing Ltd.'s pension arrangements and briefly describe their roles. [4]

ii) Outline the following risks under the DB and DC arrangements and explain how the impact differs for:

- Employee A (near retirement)
- Employee B (young member)

Risks to consider:

- Investment risk
- Longevity risk
- Inflation risk

[9]

ABC Manufacturing designs the DC scheme with:

- Voluntary employee contributions above a small mandatory minimum
- Employer matching only voluntary contributions
- Permitting partial withdrawals before retirement
- Lump sum payment at retirement (no compulsory annuitisation)
- No investment policy set so far

iii) Outline the key weaknesses of this DC design and recommend improvements, considering:

- Employer cost constraints
- Member retirement security
- Regulatory and behavioural factors

[12]

[25 Marks]

Q. 2) A textile company sponsors a closed Defined Benefit (DB) pension scheme for its employees. The scheme was established to provide retirement benefits based on a predefined formula, typically linked to salary and years of service. However, now the scheme is:

- Closed to new entrants 10 years ago
- Closed to future accrual 2 years ago
- 65% pensioners, 25% deferred members, 10% active members
- Funding level: 92% on Technical Provisions basis

- Assets: 60% return-seeking (equities, property), 40% bonds
- Employer covenant assessed as “moderate but weakening”

Trustees are considering:

- Moving to a low-risk “self-sufficiency” target
- Reducing allocation to equities
- Introducing a formal journey plan

i) Outline the main factors trustees should consider when setting a funding objective for a DB pension scheme. [3]

ii) Apply these factors to the scheme above and explain the implications for:

- Investment strategy
- Funding contributions
- Risk management

[8]

iii) The employer argues that reducing equity exposure now would:

- Increase short-term contributions
- Reduce business investment capacity

The trustees favour de-risking due to maturing liabilities.

Outline how the trustees should balance:

- Member security
- Sponsor affordability
- Long-term scheme strategy

Recommend a way forward and justify your approach.

[12]

[23 Marks]

Q. 3) XYZ company sponsors a funded Defined Benefit (DB) pension scheme.

Recent actuarial valuation results show:

- Funding level: 90%
- Salary increases over last 3 years averaged 7% p.a. (previous assumption 5%)
- Actual mortality experience lighter than expected
- Inflation volatile between 4% and 8%
- Asset returns averaged 5% vs assumed 6%

Trustees are reviewing the valuation assumptions and the actuarial projection model.

i) Outline the objectives and basic features of an actuarial projection model used for the financial management of pension benefit provision in an Indian DB scheme. [5]

ii) The trustees are concerned that the current actuarial model:

- Uses aggregated membership data

- Assumes uniform salary growth
- Does not explicitly allow for inter-dependencies between assumptions

Explain:

- The limitations of using such an aggregated deterministic model
- How you would enhance the model structure
- The likely impact of these refinements on projected liabilities and contribution rates

[10]

iii) The Indian economy has experienced significant inflation volatility (RBI target 4–6%).

Evaluate:

- The limitations of deterministic modelling
- The advantages and limitations of stochastic modelling

Recommend an appropriate sensitivity and stress-testing framework relevant to India.

Advise trustees how to use the results to set prudent but affordable contribution rates.

[12]

[27 Marks]

Q.4) ZERO Motors Ltd sponsors a defined benefit pension scheme. As, at 31 March 2026, the actuarial valuation has shown an increase in deficit of ₹23 Cr. The scheme is closed to new entrants but open to accrual.

The membership profile of the scheme is:

- 50% pensioners
- 35% deferreds
- 15% actives

Sponsor covenant:

- Recently downgraded from A- to BBB
- Profits volatile due to export exposure

The management has requested the Scheme Actuary to explain the possible reasons for the increase in deficit.

i) Discuss the various factors that may have contributed to the growth in the deficit of the scheme. [6]

The management is concerned about the increasing deficit and volatility in contribution requirements over recent years.

ii) Describe how the principles of the Actuarial Control Cycle can be applied to the assumption-setting and funding process in order to achieve a more stable contribution rate while maintaining benefit security. [7]

Observing the steadily increasing deficit, the trustees are considering:

- increasing contributions from 15% to 18%,
- revising the actuarial assumptions, and
- adopting a lower-risk investment strategy.

iii) Discuss the factors the trustees should consider before implementing these measures, highlighting how the trustees' decisions may impact other stakeholders of the pension scheme. **[12]**

[25 Marks]
